

Vera L. Casquero

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EDUCATION

University of Notre Dame – College of Engineering – Bachelor of Science in Engineering **Notre Dame, IN**
Major: Computer Science | **Minor:** Engineering Corporate Practice **May 2027**
Coursework: Data Structures (C/Python), Fundamentals of Computing, Systems Programming, Discrete Math, Linear Algebra & Differential Equations

RELEVANT WORK EXPERIENCE

Iris Recognition Research Internship | Notre Dame, Indiana **May -December 2024**
Intern

- Led the development of a **Python-based system** to classify and filter iris data, enhancing dataset accuracy for research purposes by up to **25%**.
- Co-engineered a **Python** algorithm for generating image reports and unique IDs to data, aiding synthetic iris testing and saving **50+ hours** of manual work for future research.
- Employed **deep learning ResNet-18 architecture** to evaluate a PAD system, achieving **100% accuracy** in distinguishing authentic irises images from attack images.

ACTIVITIES & PROJECTS

Student International Business Council | Boston, MA **October 2024 – December 2024**
Amazon – Travel Team Analyst

- Optimized Amazon's main page interface by enhancing underused features and collaborating with the Interface team to produce a **Figma** mockup, driving higher user engagement, retention and sales
- Conducted thorough research and analysis, projecting a **10-15% increase in retention rates** with proposed interface improvements.
- Presented a redesigned interface enhancing Generative AI features by proposing the integration of an **RCNN framework** for image classification and context-based product recognition, leveraging **deep learning** techniques like **ResNetV1** and **L-Softmax** to deliver personalized user experiences and improve small object detection accuracy.

Amazon – Project Leader **January 2025 – Present**

- Leading a team of **60+** students in technical counseling for Amazon Prime enhancements.
- Currently working on analyzing customer data using **SQL** to identify trends and insights, enabling the development of data-driven solutions for feature optimization.

Post-graduate Employment App | University of Notre Dame **Abril-May 2024**

- Developed an app comparing employment outcome for graduates across demographics focused on STEM fields, serving over **2,000+ data points** utilizing **MATLAB** frameworks.
- Created an algorithm to optimize data processing and analysis by up to **30%**, delivering faster and clear insights on how users' degrees and demographics influence employment status.

LEADERSHIP & SERVICE

Fundación Mentes Brillantes | Santo Domingo, Dominican Republic **May 2022 – Present**
Founder/President

- Established and led the first organization dedicated to teaching programming and STEM subjects to children in Dominican Republic's public school.
- Led fundraising initiatives that generated **\$50,000** in support for five public schools, directly addressing the digital divide and providing over **1,000 students** with upgraded educational resources.
- Launched a tech education program currently benefiting **65 students**, now scaling it to reach **300+ students** next year.

Notre Dame Computer Club | University of Notre Dame **January 2025 – Present**
Secretary

- Boosted club engagement by **10%** through dynamic initiatives, including speaker forums and programming workshops.
- Currently launching a competitive hackathon team to represent the club in various programming languages such as **Python, C** and **JavaScript**.

RELEVANT SKILLS

Technical: Python, C/C++, SQL, Linux, AlbaCore, SolidWorks, Microsoft Excel, 3D-Printing, MATLAB, Figma

Societies: Association of Latinos Professionals for America, Society of Women Engineers, Latino Honor Society

Language: Spanish (Fluent), English (Fluent)